

Lab-2: Discrete System Simulation Using Monte Carlo Method: Estimating Value of Pi

Lab Date: January 4th, 2025

Deadline: January 9th, 2025

Class: CSIT 5th Semester, New Summit College

Subject: Simulation & Modeling

Submission Mode: **Handwritten Document** (Print Output Section Only)

Objective:

To simulate a discrete system by estimating the value of π using the Monte Carlo simulation method.

Learning Outcomes

After completing this lab, students will be able to:

- Understand the concept of discrete systems.
- Implement the Monte Carlo method for estimation.
- Visualize the simulation process and results.
- Gain hands-on experience with random sampling and statistical estimation.

Tasks:

1. Task 1: Theoretical Understanding

- a. Research the Monte Carlo simulation method.
- b. Derive the mathematical formula for estimating π using random sampling within a square and circle.

2. Task 2: Algorithm Development

- a. Initialize the total number of iterations.
- b. Generate random (x, y) coordinates within a square.
- c. Check if the point lies inside the circle using the equation $x^2 + y^2 \leq r^2$.

- d. Count the points inside the circle and square.
- e. Estimate the value of π using the formula:

$$\pi = 4 \times ((\text{Total number of points}) / (\text{Number of points inside the circle}))$$

3. Task 3: Code Implementation

- a. Generate random (x, y) coordinates between -1 and 1.
- b. Use graphics to draw a square and circle for visualization.
- c. Use Monte Carlo simulation to estimate π and display the result.

4. Task 4: Experimentation

- a. Run the simulation for different numbers of iterations:
 - i. 100000 iterations.
 - ii. Record the estimated values of π and analyze their accuracy.
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Report Structure:

- **Objective:** Clearly state the objective of the lab.
- **Theory:** Include a brief explanation of:
 - a. Pi
 - b. Discrete System Simulation
 - c. Monte Carlo Simulation
- **Procedure:** Outline the steps to simulate the system
- **Code / Implementation**
- **Output**
- **Conclusion:** Summarize what was learned and what was observed from the simulation.

References:

1. <https://colab.research.google.com/drive/14-eJs8ft3EPHANpAt0Zm46kPYz8aLy9j?usp=sharing>

2. <https://drive.google.com/file/d/1OgmZ4RcQOmKhIN3CZliyHta9S8lvXnb4/view?usp=sharing>